A Nonlinear Control Strategy for Dual-Drug Targeting of the Kallikrein-Kinin-Complement Pathway in COVID-19

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Abstract

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection can involve dysregulated inflammatory responses associated with the kallikrein–kinin system (KKS) and the terminal complement pathway. This study develops a system of ordinary differential equations (ODEs) to model these pathways and investigates their regulation using a double-integral sliding mode controller (DISMC). Hypothetical control inputs associated with icatibant, a bradykinin B2 receptor (B2R) antagonist, and vilobelimab (IFX-1), an anti-C5a monoclonal antibody, are incorporated to examine the simulated regulation of selected model states. A genetic algorithm (GA) is employed to optimise the controller parameters, while Lyapunov analysis establishes the nominal reaching behaviour of the sliding variables. Controlled and uncontrolled simulations examine the dynamics of bradykinin (BK), the BK–B2R complex, the C5a–C5aR1 complex, and interleukin-6 (IL-6), alongside perturbations in selected initial-state values. Under the simulated conditions, control reduces the steady-state IL-6 response by approximately 15.8% relative to the uncontrolled case. Variance-based global sensitivity analysis using Sobol' indices further identifies influential initial-state

values and kinetic parameters associated with variability in the selected outputs. The drug-associated model states track their prescribed reference values, demonstrating reference-tracking behaviour within the adopted mathematical framework. These findings provide a computational proof of concept for investigating nonlinear control of coupled inflammatory pathways. However, the model states are not calibrated against experimentally measured absolute biochemical concentrations, and the control inputs and reference values do not represent clinically validated drug doses. Experimental calibration, independent validation, and further assessment of system stability, uncertainty, and robustness are required before biological predictions or therapeutic implications can be established.

Keywords: Kallikrein–kinin system; terminal complement pathway; double-integral sliding mode control; genetic algorithm; global sensitivity analysis

1 Introduction

Coronavirus disease 2019 (COVID-19), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), can involve dysregulated immune responses and excessive inflammation, contributing to tissue damage and disease severity [1, 2]. Among the inflammatory pathways implicated in COVID-19 are the kallikrein–kinin system (KKS) and the terminal complement pathway. Complement component C5a and its receptor C5aR1 participate in inflammatory signalling [3, 4], while bradykinin (BK) and its receptor B2R are associated with kallikrein–kinin pathway activity. These pathways may contribute to the inflammatory environment associated with severe infection. Interleukin-6 (IL-6), an important pro-inflammatory cytokine, is also implicated in COVID-19-associated inflammation [5, 6]. These biological relationships motivate mathematical investigations of the interactions among BK, B2R, C5a, C5aR1, and IL-6. Pharmacological agents such as icatibant, a bradykinin B2 receptor antagonist, and vilobelimab (IFX-1), an anti-C5a monoclonal antibody, provide relevant therapeutic contexts for investigating the regulation of these pathways [5, 7].

Mathematical modelling and nonlinear control offer complementary approaches for investigating complex biological systems. Ordinary differential equation (ODE) models can represent interactions among biochemical species and describe their time-dependent behaviour under specified assumptions. Nonlinear control methods can be incorporated into these models to investigate how selected system states respond to prescribed control inputs. In particular, sliding mode control (SMC) and its extensions have been applied to nonlinear dynamical systems because of their potential to regulate system trajectories through appropriately designed sliding surfaces and reaching laws. Islam et al. [8] investigated integral and double-integral sliding mode control strategies for suppressing leukemic cell proliferation and examining chemotherapy-related dynamics. Similarly, Qaiser et al. [9] explored synergistic control frameworks and state-feedback methodologies for regulating malignant tumour growth in nonlinear models. More broadly, control-theoretic approaches have been investigated in systems and synthetic biology to analyse and regulate biological processes [10–13]. These studies provide methodological foundations for applying nonlinear control to biological

systems. However, the properties established for a particular controller depend on its mathematical formulation, the system dynamics, and the assumptions underlying its analysis.

Computational modelling of the complement system has been investigated in several biological contexts. For example, mathematical models have been developed to study complement responses to pathogens such as *Neisseria meningitidis* [14]. Quantitative systems pharmacology modelling has also been used to investigate potential drug targets for reducing complement activation in autoimmune diseases [15]. In our previous study [4], we developed a kinetic discrete model to investigate the responses of complement components, including C3 and C5a, in SARS-CoV-2-associated inflammatory pathways. In related work, we developed a systems-level mathematical model of the hemostatic and complement systems to investigate interactions among coagulation, fibrinolytic, and complement components in the context of immunothrombosis and hyperinflammation [1]. These studies provide a foundation for further computational investigation of inflammatory processes involving complement activation. Nevertheless, investigating the coupled dynamics of the KKS and terminal complement pathway within a nonlinear control framework, while examining selected pathway states and their relationship with IL-6, presents a distinct modelling and analysis problem.

In this study, we investigate a mathematical model of the KKS and terminal complement inflammatory pathway using a double-integral sliding mode controller (DISMC). Hypothetical control inputs associated with icatibant and vilobelimab (IFX-1) are incorporated to examine the simulated regulation of selected model states. A genetic algorithm (GA) is used to optimise the controller parameters according to the specified objective function. The analysis examines the nominal reaching behaviour of the sliding variables using Lyapunov analysis and evaluates the simulated dynamics of BK, the BK–B2R complex, the C5a–C5aR1 complex, and IL-6. The drug-associated model states are also examined in relation to prescribed reference values (set points). In addition, variance-based global sensitivity analysis (GSA), using the Sobol' method, is performed to quantify the contributions of selected model inputs to the variability of the chosen outputs. Simulations involving perturbations in selected initial-state values are used to examine the sensitivity of the model trajectories to changes in the initial conditions.

The principal contribution of this study is a computational proof of concept for investigating the controlled dynamics of the coupled KKS–terminal complement model and its associated IL-6 response. The numerical simulations compare controlled and uncontrolled model trajectories, examine reference tracking for drug-associated states, and quantify the sensitivity of selected outputs to variations in model inputs. The interpretation of these findings is restricted to the mathematical model, parameter choices, and simulation assumptions adopted in this study. In particular, the model states are not calibrated against experimentally measured absolute biochemical concentrations, and the control inputs and reference values do not represent clinically established drug doses. Furthermore, the Lyapunov analysis addresses the nominal reaching behaviour of the sliding variables; it does not establish stability of the complete biochemical system or robustness against external disturbances and parameter

uncertainties. Chattering reduction is likewise not established by the controller formulation analysed here. These limitations define the scope of the present computational investigation and the need for further validation.

The remainder of this paper is organised as follows. Section 2 presents the materials and methods, including the mathematical formulation of the model. Section 3 describes the design of the double-integral sliding mode controller (DISMC), including its formulation, nominal reaching-law analysis, and genetic algorithm (GA)-based parameter optimisation. Section 4 presents the variance-based global sensitivity analysis (GSA) using the Sobol' method, including the mathematical formulations of the first-order and total-order sensitivity indices. Section 5 presents the numerical simulation results, including the time-course profiles of the model state variables and the GSA findings. Finally, Sections 6 and 7 present the discussion of the results and the conclusions of the study, respectively.

2 Materials and Methods

The mathematical model comprises a system of ordinary differential equations (ODEs) developed in MATLAB SimBiology, while MATLAB Simulink is used to implement the nonlinear double-integral sliding mode controller (DISMC). The model incorporates the kinetic parameters listed in Supplementary Table S1, with values obtained from the relevant literature. Microarray gene-expression data [1] provide the biological context for model development; however, these data are not directly converted into protein concentrations. The state variables are assigned numerical values for simulation purposes and are therefore interpreted as model-defined abundances rather than experimentally calibrated concentrations. Although the original formulation expressed these values in molarity (M), their correspondence to experimentally measured physical concentrations has not been established. The model state variables and their assigned values are listed in Table 1. Numerical simulations are performed to investigate the temporal dynamics of the model variables and the effects of the control inputs, while variance-based global sensitivity analysis (GSA) is used to quantify the contributions of model parameters to variations in the selected outputs. The reference values assigned to icatibant and vilobelimab serve as model-defined control set points rather than physical drug concentrations or clinically applicable doses.

2.1 Mathematical model formalism

As shown in Fig. 1, the biochemical reaction network is described by a set of state variables representing the main biological molecules, receptors, and their complexes, as well as the control agents. These variables are defined as follows: x_1 represents bradykinin (BK); x_2, the bradykinin B2 receptor (B2R); x_3, the BK–B2R complex; x_4, activated thrombin-activatable fibrinolysis inhibitor (TAFIa); x_5, prekallikrein (P-KAL); x_6, kallikrein (KAL); x_7, high-molecular-weight kininogen (HK); x_8, the P-KAL–HK complex; x_9, interleukin-6 (IL-6); x_{10}, icatibant; x_{11}, complement component C5; x_{12}, C5a; x_{13}, C5b; x_{14}, C5a receptor 1 (C5aR1, also known as CD88);

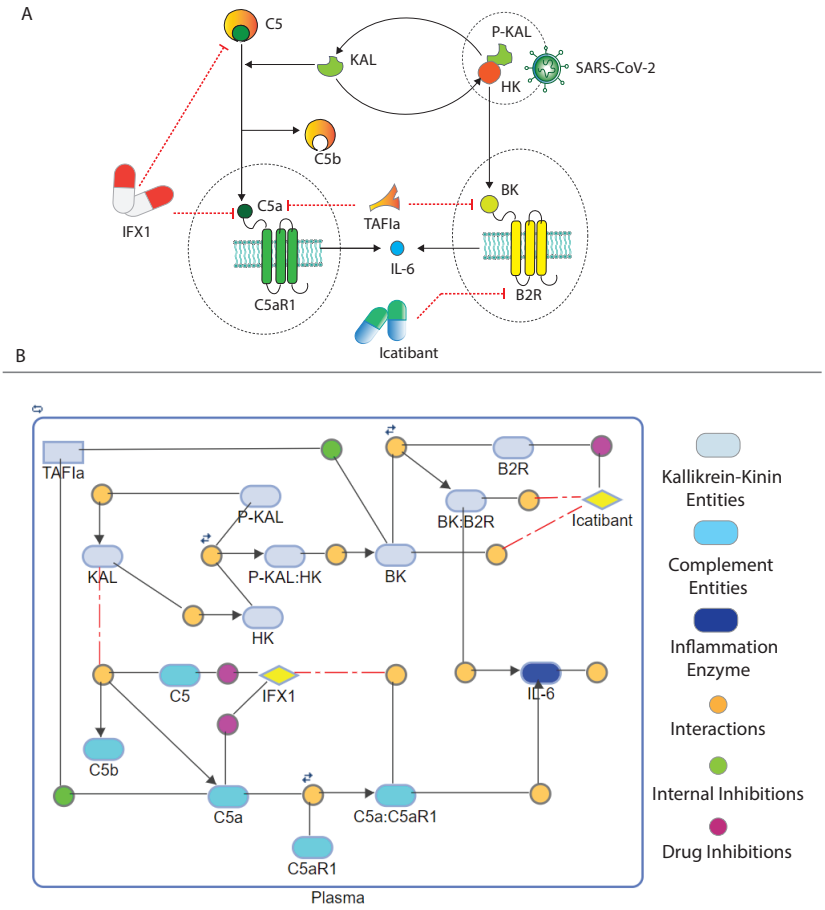


Fig. 1: (A) Schematic representation of the kallikrein–kinin and complement pathways, illustrating the formation of the BK–B2R and C5a–C5aR1 complexes and their association with interleukin-6 (IL-6) production. The pre-kallikrein (P-KAL)–high-molecular-weight kininogen (HK) complex participates in kallikrein–kinin pathway activation [16]. Kallikrein (KAL) can cleave complement component C5, generating C5a and C5b [17, 18]. Activated thrombin-activatable fibrinolysis inhibitor (TAFIa) inhibits the activities of bradykinin (BK) [19] and C5a. (B) Biochemical reaction network model developed using MATLAB SimBiology.

x_{15}, the C5a–C5aR1 complex; and x_{16}, IFX-1 (vilobelimab). The dynamics and interactions of these variables are mathematically described by the system of ordinary differential equations (ODEs) given in Eq. 1.

$$
\begin{aligned}
\frac{dx_1}{dt} &= -k_\alpha^+ x_1 x_2 + k_\alpha^- x_3 + k_\alpha x_8 - k_{\beta 1}^+ x_1 x_4 - \frac{k_{c1} x_1 x_{10}}{k_{m1} + x_{10}} \\
\frac{dx_2}{dt} &= -k_\alpha^+ x_1 x_2 + k_\alpha^- x_3 - k_{\gamma 1}^+ x_{10} x_2 \\
\frac{dx_3}{dt} &= k_\alpha^+ x_1 x_2 - k_\alpha^- x_3 - k_{\omega 1} x_3 - \frac{k_{c2} x_3 x_{10}}{k_{m2} + x_{10}} \\
\frac{dx_4}{dt} &= -k_{\beta 1}^+ x_1 x_4 - k_{\beta 2}^+ x_4 x_{12} \\
\frac{dx_5}{dt} &= -k_\gamma^+ x_5 x_6 - k_\tau^+ x_5 x_7 + k_\tau^- x_8 \\
\frac{dx_6}{dt} &= k_\gamma^+ x_5 x_6 - k_\sigma^+ x_6 x_7 \\
\frac{dx_7}{dt} &= -k_\tau^+ x_5 x_7 + k_\tau^- x_8 + k_\sigma^+ x_6 x_7 \\
\frac{dx_8}{dt} &= k_\tau^+ x_5 x_7 - k_\tau^- x_8 - k_\alpha x_8 \\
\frac{dx_9}{dt} &= k_{\omega 1} x_3 + k_{\omega 2} x_{15} - k_\varphi^- x_9 \\
\frac{dx_{10}}{dt} &= -k_{\gamma 1}^+ x_{10} x_2 + u_1(t) \\
\frac{d(x_{11})}{dt} &= -\frac{k_{c3} x_6 x_{11}}{k_{m3} + x_6} - k_{\gamma 2}^+ x_{11} x_{16} \\
\frac{d(x_{12})}{dt} &= \frac{k_{c3} x_6 x_{11}}{k_{m3} + x_6} - k_\lambda^+ x_{12} x_{14} + k_\lambda^- x_{15} \\
&\quad - k_{\gamma 3}^+ x_{12} x_{16} - k_{\beta 2}^+ x_4 x_{12} \\
\frac{d(x_{13})}{dt} &= \frac{k_{c3} x_6 x_{11}}{k_{m3} + x_6} \\
\frac{d(x_{14})}{dt} &= -k_\lambda^+ x_{12} x_{14} + k_\lambda^- x_{15} \\
\frac{d(x_{15})}{dt} &= k_\lambda^+ x_{12} x_{14} - k_\lambda^- x_{15} - k_{\omega 2} x_{15} - \frac{k_{c4} x_{16} x_{15}}{k_{m4} + x_{16}} \\
\frac{d(x_{16})}{dt} &= -k_{\gamma 2}^+ x_{11} x_{16} - k_{\gamma 3}^+ x_{12} x_{16} + u_2(t).
\end{aligned}
\tag{1}
$$

The control inputs $u_1(t)$ and $u_2(t)$ represent hypothetical mathematical control signals associated with icatibant and vilobelimab, respectively, rather than clinically established drug doses. Icatibant is a selective bradykinin B2 receptor (B2R) antagonist that inhibits BK-mediated receptor activation [5], whereas vilobelimab is an anti-C5a monoclonal antibody that binds C5a and inhibits its biological activity [7]. The genetic algorithm optimises the controller parameters to improve simulated tracking performance within the adopted mathematical formulation; it does not establish clinically validated dosing regimens. Determining clinically applicable doses would require pharmacokinetic/pharmacodynamic modelling, clinically meaningful exposure constraints, and experimental or clinical validation. The mathematical formulations of $u_1(t)$ and $u_2(t)$ are given in Eqs. (17) and (25), respectively.

3 Controller design

3.1 Sliding mode controller

The sliding mode controller (SMC) is a nonlinear control strategy that drives a system toward a predefined sliding surface determined by the tracking errors between the actual and desired states. Under appropriate control-design assumptions, SMC can provide rapid convergence and robustness to certain bounded disturbances and model uncertainties. However, these properties depend on the system dynamics, the control law, and the assumptions used in the stability analysis. A common limitation of conventional SMC is chattering, which refers to high-frequency oscillations associated with switching control action near the sliding surface. The presence and severity of chattering depend on the controller formulation, system dynamics, sampling resolution, and implementation characteristics.

In this study, a double-integral sliding mode controller (DISMC) is formulated by incorporating tracking errors and their integral and double-integral terms into the sliding surfaces. These terms provide additional flexibility in shaping the tracking dynamics. The simulated bradykinin (BK) trajectory exhibits an irregular transient response following icatibant intervention; however, this behaviour alone does not establish the presence of high-frequency chattering. Similarly, the simulated biological state trajectories do not provide sufficient evidence to establish sustained high-frequency oscillations. The control inputs $u_1(t)$ and $u_2(t)$ correspond to icatibant and vilobelimab, respectively, and enter the drug-state equations associated with x_{10} and x_{16}. Although the controller is designed to drive the sliding variables toward zero, the chattering-reduction capability of the proposed DISMC has not been established through a dedicated quantitative analysis of the control-input trajectories. Therefore, no explicit claim of guaranteed chattering reduction is made in this study. The controller formulation and its nominal reaching behaviour are described below.

3.2 Double Integral Sliding Mode Controller

Consider the first sliding surface defined as

$$\begin{aligned} S_1 = {} & c_1e_1 + c_2e_2 + c_3e_3 + c_4e_4 \\ & + c_5e_{1a} + c_6e_{2a} + c_7e_{3a} + c_8e_{4a} \\ & + c_9e_{1b} + c_{10}e_{2b} + c_{11}e_{3b} + c_{12}e_{4b}. \end{aligned} \tag{2}$$

Here, $c_i \in \mathbb{R}^+$, for $i = 1, \ldots, 12$, are the sliding-surface coefficients.

The tracking errors are defined as

$$\begin{aligned} e_1 &= x_1 - x_{1ref}, \\ e_2 &= x_2 - x_{2ref}, \\ e_3 &= x_3 - x_{3ref}, \\ e_4 &= x_{10} - x_{10ref}. \end{aligned} \tag{3}$$

The corresponding integral error terms are

$$\begin{aligned} e_{1a} &= \int (x_1 - x_{1ref})\, dt, \\ e_{2a} &= \int (x_2 - x_{2ref})\, dt, \\ e_{3a} &= \int (x_3 - x_{3ref})\, dt, \\ e_{4a} &= \int (x_{10} - x_{10ref})\, dt. \end{aligned} \tag{4}$$

Integrating these error terms once more gives the double-integral error terms:

$$\begin{aligned} e_{1b} &= \int e_{1a}\, dt, \\ e_{2b} &= \int e_{2a}\, dt, \\ e_{3b} &= \int e_{3a}\, dt, \\ e_{4b} &= \int e_{4a}\, dt. \end{aligned} \tag{5}$$

Differentiating the sliding surface in Eq. (2) yields

$$\begin{aligned} \dot{S}_1 = {} & c_1\dot{e}_1 + c_2\dot{e}_2 + c_3\dot{e}_3 + c_4\dot{e}_4 \\ & + c_5\dot{e}_{1a} + c_6\dot{e}_{2a} + c_7\dot{e}_{3a} + c_8\dot{e}_{4a} \\ & + c_9\dot{e}_{1b} + c_{10}\dot{e}_{2b} + c_{11}\dot{e}_{3b} + c_{12}\dot{e}_{4b}. \end{aligned} \tag{6}$$

The derivatives of the tracking errors are

$$\begin{aligned} \dot{e}_1 &= \dot{x}_1 - \dot{x}_{1ref}, \\ \dot{e}_2 &= \dot{x}_2 - \dot{x}_{2ref}, \\ \dot{e}_3 &= \dot{x}_3 - \dot{x}_{3ref}, \\ \dot{e}_4 &= \dot{x}_{10} - \dot{x}_{10ref}. \end{aligned} \tag{7}$$

The derivatives of the integral error terms satisfy

$$\begin{aligned} \dot{e}_{1a} &= e_1, \\ \dot{e}_{2a} &= e_2, \\ \dot{e}_{3a} &= e_3, \\ \dot{e}_{4a} &= e_4. \end{aligned} \tag{8}$$

Similarly, the derivatives of the double-integral error terms are

$$\begin{aligned} \dot{e}_{1b} &= e_{1a}, \\ \dot{e}_{2b} &= e_{2a}, \\ \dot{e}_{3b} &= e_{3a}, \\ \dot{e}_{4b} &= e_{4a}. \end{aligned} \tag{9}$$

Substituting Eqs. (7)–(9) into Eq. (6) gives

$$\begin{aligned}\dot{S}_1 = {} & c_1(\dot{x}_1 - \dot{x}_{1ref}) + c_2(\dot{x}_2 - \dot{x}_{2ref}) \\ & + c_3(\dot{x}_3 - \dot{x}_{3ref}) + c_4(\dot{x}_{10} - \dot{x}_{10ref}) \\ & + c_5 e_1 + c_6 e_2 + c_7 e_3 + c_8 e_4 \\ & + c_9 e_{1a} + c_{10} e_{2a} + c_{11} e_{3a} + c_{12} e_{4a}.\end{aligned} \tag{10}$$

3.2.1 Nominal reaching law and Lyapunov analysis

To specify the desired reaching behaviour, the following nominal reaching law is adopted, following the general formulation of sliding-mode reaching laws [20]:

$$\dot{S}_i = -k_i |S_i|^{\alpha_i} \operatorname{sign}(S_i), \qquad i = 1, 2, \tag{11}$$

where $k_i > 0$ is a design coefficient and $0 \leq \alpha_i < 1$ is the reaching-law exponent. The signum function is defined as

$$\operatorname{sign}(S_i) = \begin{cases} 1, & S_i > 0, \\ 0, & S_i = 0, \\ -1, & S_i < 0. \end{cases}$$

The original expression $\operatorname{sign}(S_i/\beta_i)$ has been written as $\operatorname{sign}(S_i)$ because, for any $\beta_i > 0$,

$$\operatorname{sign}\left(\frac{S_i}{\beta_i}\right) = \operatorname{sign}(S_i).$$

Consequently, scaling the argument of the signum function by a positive constant does not smooth its discontinuity or, by itself, mitigate chattering. The parameter β_i is therefore omitted from the reaching law, and no chattering-reduction property is attributed to it.

Each sliding surface combines multiple tracking errors with their integral and double-integral terms. Consequently, the reaching law regulates the associated combined sliding variable, while the coefficients determine the relative contributions of the individual error terms. Convergence of a sliding variable to zero does not, by itself, guarantee convergence of every individual tracking error. Such convergence depends on the reduced-order dynamics on the sliding surface and the selected coefficients and reference trajectories.

3.2.2 Theorem

Consider the nominal sliding-variable dynamics governed by Eq. (11), with $k_i > 0$ and $0 \leq \alpha_i < 1$. For the Lyapunov candidate

$$\theta_i = \frac{1}{2} S_i^2, \tag{12}$$

the derivative satisfies

$$\dot{\theta}_i = S_i \dot{S}_i. \tag{13}$$

Substituting Eq. (11) into Eq. (13) gives

$$\begin{aligned} \dot{\theta}_i &= -k_i S_i |S_i|^{\alpha_i} \operatorname{sign}(S_i) \\ &= -k_i |S_i|^{\alpha_i + 1} \leq 0. \end{aligned} \tag{14}$$

Therefore, θ_i is non-increasing and strictly decreases whenever $S_i \neq 0$.

Proof. The Lyapunov candidate in Eq. (12) is positive definite with respect to S_i and satisfies $\theta_i = 0$ if and only if $S_i = 0$. Differentiating it gives Eq. (13). Using the reaching law in Eq. (11), we obtain

$$\begin{aligned} \dot{\theta}_i &= S_i \left[-k_i |S_i|^{\alpha_i} \operatorname{sign}(S_i) \right] \\ &= -k_i |S_i|^{\alpha_i + 1} \leq 0. \end{aligned}$$

Under the specified nominal reaching law, the sliding variable reaches zero in finite time. This result establishes the reaching behaviour of the sliding variable under the assumed dynamics; it does not, by itself, establish stability of the complete biochemical system or robustness to external disturbances and parameter variations.

3.2.3 Derivation of the first control input

Equating Eq. (10) with the reaching law in Eq. (11) gives

$$\begin{aligned} -k_1 |S_1|^{\alpha_1} \operatorname{sign}(S_1) = {} & c_1(\dot{x}_1 - \dot{x}_{1ref}) \\ & + c_2(\dot{x}_2 - \dot{x}_{2ref}) \\ & + c_3(\dot{x}_3 - \dot{x}_{3ref}) \\ & + c_4(\dot{x}_{10} - \dot{x}_{10ref}) \\ & + c_5 e_1 + c_6 e_2 + c_7 e_3 + c_8 e_4 \\ & + c_9 e_{1a} + c_{10} e_{2a} + c_{11} e_{3a} + c_{12} e_{4a}. \end{aligned} \tag{15}$$

Substituting the state equations from Eq. (1) into Eq. (15) gives

$$\begin{aligned}
-k_1|S_1|^{\alpha_1}\,\mathrm{sign}(S_1) = c_1\Big(&-k_\alpha^+ x_1x_2 + k_\alpha^- x_3 + k_\alpha x_8 \\
&-k_{\beta1}^+ x_1x_4 - \frac{k_{c1}x_1x_{10}}{k_{m1}+x_{10}} - \dot{x}_{1ref}\Big) \\
&+ c_2\left(-k_\alpha^+ x_1x_2 + k_\alpha^- x_3 - k_{\gamma1}^+ x_{10}x_2 - \dot{x}_{2ref}\right) \\
&+ c_3\left(k_\alpha^+ x_1x_2 - k_\alpha^- x_3 - k_{\omega1}x_3 - \frac{k_{c2}x_3x_{10}}{k_{m2}+x_{10}} - \dot{x}_{3ref}\right) \\
&+ c_4\left(-k_{\gamma1}^+ x_{10}x_2 + u_1(t) - \dot{x}_{10ref}\right) \\
&+ c_5e_1 + c_6e_2 + c_7e_3 + c_8e_4 \\
&+ c_9e_{1a} + c_{10}e_{2a} + c_{11}e_{3a} + c_{12}e_{4a}.
\end{aligned} \tag{16}$$

Rearranging Eq. (16) to isolate $u_1(t)$ yields

$$\begin{aligned}
u_1(t) = &-\frac{k_1}{c_4}|S_1|^{\alpha_1}\,\mathrm{sign}(S_1) \\
&-\frac{c_1}{c_4}\Big(-k_\alpha^+ x_1x_2 + k_\alpha^- x_3 + k_\alpha x_8 \\
&\quad -k_{\beta1}^+ x_1x_4 - \frac{k_{c1}x_1x_{10}}{k_{m1}+x_{10}} - \dot{x}_{1ref}\Big) \\
&-\frac{c_2}{c_4}\left(-k_\alpha^+ x_1x_2 + k_\alpha^- x_3 - k_{\gamma1}^+ x_{10}x_2 - \dot{x}_{2ref}\right) \\
&-\frac{c_3}{c_4}\left(k_\alpha^+ x_1x_2 - k_\alpha^- x_3 - k_{\omega1}x_3 - \frac{k_{c2}x_3x_{10}}{k_{m2}+x_{10}} - \dot{x}_{3ref}\right) \\
&+ k_{\gamma1}^+ x_{10}x_2 + \dot{x}_{10ref} \\
&-\frac{c_5}{c_4}e_1 - \frac{c_6}{c_4}e_2 - \frac{c_7}{c_4}e_3 - \frac{c_8}{c_4}e_4 \\
&-\frac{c_9}{c_4}e_{1a} - \frac{c_{10}}{c_4}e_{2a} - \frac{c_{11}}{c_4}e_{3a} - \frac{c_{12}}{c_4}e_{4a}.
\end{aligned} \tag{17}$$

3.2.4 Derivation of the second control input

The second sliding surface is defined as

$$\begin{aligned}
S_2 = &\, d_1e_5 + d_2e_6 + d_3e_7 + d_4e_8 \\
&+ d_5e_{5a} + d_6e_{6a} + d_7e_{7a} + d_8e_{8a} \\
&+ d_9e_{5b} + d_{10}e_{6b} + d_{11}e_{7b} + d_{12}e_{8b}.
\end{aligned} \tag{18}$$

Here, $d_i \in \mathbb{R}^+$, for $i = 1, \ldots, 12$, are the sliding-surface coefficients.

The tracking errors are defined as

$$\begin{aligned} e_5 &= x_{11} - x_{11ref}, \\ e_6 &= x_{12} - x_{12ref}, \\ e_7 &= x_{15} - x_{15ref}, \\ e_8 &= x_{16} - x_{16ref}. \end{aligned} \tag{19}$$

The corresponding integral error terms are

$$\begin{aligned} e_{5a} &= \int e_5 \, dt, \quad e_{6a} = \int e_6 \, dt, \\ e_{7a} &= \int e_7 \, dt, \quad e_{8a} = \int e_8 \, dt. \end{aligned} \tag{20}$$

The double-integral error terms are

$$\begin{aligned} e_{5b} &= \int e_{5a} \, dt, \quad e_{6b} = \int e_{6a} \, dt, \\ e_{7b} &= \int e_{7a} \, dt, \quad e_{8b} = \int e_{8a} \, dt. \end{aligned} \tag{21}$$

Differentiating Eq. (18) gives

$$\begin{aligned} \dot{S}_2 = {} & d_1\dot{e}_5 + d_2\dot{e}_6 + d_3\dot{e}_7 + d_4\dot{e}_8 \\ & + d_5\dot{e}_{5a} + d_6\dot{e}_{6a} + d_7\dot{e}_{7a} + d_8\dot{e}_{8a} \\ & + d_9\dot{e}_{5b} + d_{10}\dot{e}_{6b} + d_{11}\dot{e}_{7b} + d_{12}\dot{e}_{8b}. \end{aligned} \tag{22}$$

Using the reaching law in Eq. (11) and the derivatives of the tracking, integral, and double-integral error terms, we obtain

$$\begin{aligned} -k_2|S_2|^{\alpha_2}\,\mathrm{sign}(S_2) = {} & d_1(\dot{x}_{11} - \dot{x}_{11ref}) \\ & + d_2(\dot{x}_{12} - \dot{x}_{12ref}) \\ & + d_3(\dot{x}_{15} - \dot{x}_{15ref}) \\ & + d_4(\dot{x}_{16} - \dot{x}_{16ref}) \\ & + d_5e_5 + d_6e_6 + d_7e_7 + d_8e_8 \\ & + d_9e_{5a} + d_{10}e_{6a} + d_{11}e_{7a} + d_{12}e_{8a}. \end{aligned} \tag{23}$$

Substituting the state equations from Eq. (1) into Eq. (23) gives

$$
\begin{aligned}
-k_2|S_2|^{\alpha_2}\,\mathrm{sign}(S_2) = d_1 &\left(-\frac{k_{c3}x_6x_{11}}{k_{m3}+x_6} - k^+_{\gamma 2}x_{11}x_{16} - \dot{x}_{11ref}\right) \\
&+ d_2\left(\frac{k_{c3}x_6x_{11}}{k_{m3}+x_6} - k^+_{\lambda}x_{12}x_{14} + k^-_{\lambda}x_{15}\right. \\
&\quad \left. -k^+_{\gamma 3}x_{12}x_{16} - k^+_{\beta 2}x_4x_{12} - \dot{x}_{12ref}\right) \\
&+ d_3\left(k^+_{\lambda}x_{12}x_{14} - k^-_{\lambda}x_{15} - k_{\omega 2}x_{15}\right. \\
&\quad \left. -\frac{k_{c4}x_{16}x_{15}}{k_{m4}+x_{16}} - \dot{x}_{15ref}\right) \\
&+ d_4\left(-k^+_{\gamma 2}x_{11}x_{16} - k^+_{\gamma 3}x_{12}x_{16} + u_2(t) - \dot{x}_{16ref}\right) \\
&+ d_5e_5 + d_6e_6 + d_7e_7 + d_8e_8 \\
&+ d_9e_{5a} + d_{10}e_{6a} + d_{11}e_{7a} + d_{12}e_{8a}.
\end{aligned}
\tag{24}
$$

Rearranging Eq. (24) to isolate $u_2(t)$ yields

$$
\begin{aligned}
u_2(t) = &-\frac{k_2}{d_4}|S_2|^{\alpha_2}\,\mathrm{sign}(S_2) \\
&-\frac{d_1}{d_4}\left(-\frac{k_{c3}x_6x_{11}}{k_{m3}+x_6} - k^+_{\gamma 2}x_{11}x_{16} - \dot{x}_{11ref}\right) \\
&-\frac{d_2}{d_4}\left(\frac{k_{c3}x_6x_{11}}{k_{m3}+x_6} - k^+_{\lambda}x_{12}x_{14} + k^-_{\lambda}x_{15}\right. \\
&\quad \left. -k^+_{\gamma 3}x_{12}x_{16} - k^+_{\beta 2}x_4x_{12} - \dot{x}_{12ref}\right) \\
&-\frac{d_3}{d_4}\left(k^+_{\lambda}x_{12}x_{14} - k^-_{\lambda}x_{15} - k_{\omega 2}x_{15}\right. \\
&\quad \left. -\frac{k_{c4}x_{16}x_{15}}{k_{m4}+x_{16}} - \dot{x}_{15ref}\right) \\
&+ k^+_{\gamma 2}x_{11}x_{16} + k^+_{\gamma 3}x_{12}x_{16} + \dot{x}_{16ref} \\
&-\frac{d_5}{d_4}e_5 - \frac{d_6}{d_4}e_6 - \frac{d_7}{d_4}e_7 - \frac{d_8}{d_4}e_8 \\
&-\frac{d_9}{d_4}e_{5a} - \frac{d_{10}}{d_4}e_{6a} - \frac{d_{11}}{d_4}e_{7a} - \frac{d_{12}}{d_4}e_{8a}.
\end{aligned}
\tag{25}
$$

Equations (17) and (25) define the nominal control laws for $u_1(t)$ and $u_2(t)$, corresponding to icatibant and vilobelimab, respectively. These control laws are derived by equating the derivatives of the respective sliding surfaces to the nominal reaching law and isolating the corresponding control inputs. The Lyapunov analysis establishes the reaching behaviour of the sliding variables under the specified nominal dynamics. However, convergence of the sliding variables does not necessarily imply convergence

of every individual tracking error or stability of the complete biochemical system. Furthermore, because external disturbances and parameter uncertainties have not been explicitly incorporated into the reaching-law analysis with defined bounds, robustness to these effects is not established by the present theorem. Quantitative assessment of chattering and a formal robustness analysis remain subjects for further investigation.

The reference trajectories define the desired model states used to construct the tracking errors in the two sliding surfaces. All eight reference trajectories are treated as constant in the numerical simulations. The reference values for the drug-associated states are $(x_{10}, \mathrm{ref}(t) = 2)$ and $(x_{16}, \mathrm{ref}(t) = 3)$, representing model-defined numerical set points rather than experimentally calibrated drug concentrations or clinically validated therapeutic targets. The reference values for the biological states x_1, x_2, x_3, x_{11}, x_{12}, and x_{15} are those specified in the numerical implementation and are selected for the mathematical tracking analysis. These values are not intended to represent experimentally established physiological targets.

3.3 Genetic Algorithm

The genetic algorithm is a widely adopted optimization technique used to address complex optimization problems. It operates by simulating natural selection and evolution, where it iteratively selects the fittest individuals from a population and generates new offspring that inherit desirable traits from their parents. In the context of optimizing the tuning parameters for DISMC, GA employs specific parameters: a population size of 50, a crossover rate of 0.8, a mutation rate of 0.1, and a maximum of 100 iterations. During each iteration, the GA algorithm assesses the fitness of individuals, applies selection, crossover, and mutation operations to produce new offspring, and replaces the least fit individuals accordingly. This iterative process continues until a stopping criterion, such as reaching the maximum number of iterations or achieving a desired level of convergence, is met. Fig. 2 illustrates the flow chart detailing the GA algorithm. The GA employs the following operators:

- Selection operator: The algorithm utilizes the Roulette wheel selection method to prioritize individuals with higher fitness values for reproduction.
- Crossover operator: New offspring are generated using the Uniform crossover method, randomly selecting tuning parameters from the parents.
- Mutation operator: The algorithm employs Gaussian mutation to introduce slight random adjustments in the tuning parameters of the offspring.

The cost function C_i, which plays a crucial role in minimizing the error term, is defined as follows:

$$min\, C_i(S) = min \int_1^t \tau |S_i| d\tau \tag{26}$$

Because the GA is a stochastic, population-based optimisation method, its convergence depends on the search-space characteristics, population diversity, and optimisation settings. Although roulette-wheel selection, uniform crossover, and Gaussian mutation facilitate the exploration and refinement of candidate solutions, premature

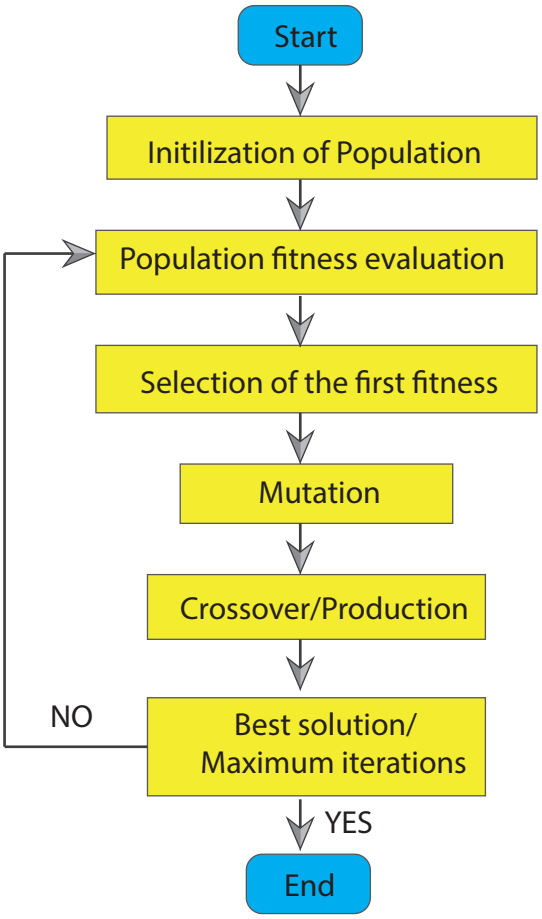


Fig. 2: Flowchart of genetic algorithm.

convergence to a local optimum cannot be ruled out. Accordingly, the obtained parameter set is regarded as a near-optimal solution under the specified optimisation settings, rather than a mathematically guaranteed global optimum.

The optimized parameters for the control inputs u_1 and u_2 obtained using the GA and used in the simulations are listed below.

For u_1: $c_1 = 0.1$, $c_2 = 0.01$, $c_3 = 0.001$, $c_4 = 10^{11}$, $c_5 = \cdots = c_{12} = 0.1$, $\alpha_1 = 0.95$, $\beta_1 = 0.95$, and $k_1 = 0.1$.

For u_2: $d_1 = 0.01$, $d_2 = 0.001$, $d_3 = 0.0001$, $d_4 = 10^{14}$, $d_5 = \cdots = d_{12} = 0.1$, $\alpha_2 = 0.95$, $\beta_2 = 0.95$, and $k_2 = 0.1$.

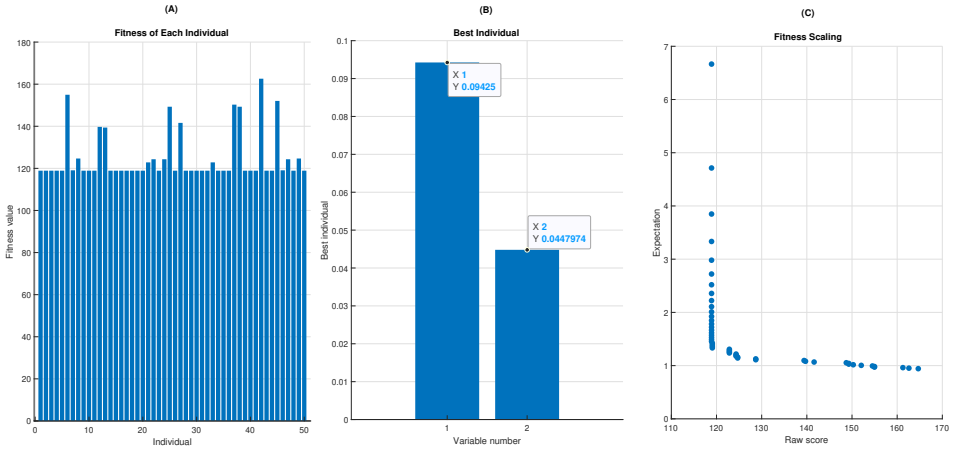


Fig. 3: (A) The fitness of individuals in the population are assessed via a fitness function. This evaluation criterion measures how effectively each solution, defined by a specific set of gain values, impacts system behavior. The fitness function considers transient response characteristics, steady-state errors, and relevant performance metrics intrinsic to the system's dynamics. (B) The term current best individual refers to the particular solution or configuration of gain values that achieves the highest fitness score or the most optimal control performance within the current population. During each iteration, fitness scores are compared, and the individual with the highest score is identified as the optimal choice. (C) The fitness scaling process seeks to adjust the fitness scores assigned to individuals, ensuring a fair distribution that supports effective selection in subsequent stages. This adjustment is especially useful when initial fitness scores show limited variability or significant skewness. By refining the scores, the process enhances the chances of a diverse range of individuals being chosen for advancement to the next generation.

4 Sensitivity analysis

Sensitivity analysis (SA) evaluates how uncertainty in model inputs, including initial state values and kinetic parameters, influences model outputs. It helps identify influential inputs and assess the reliability of model predictions [21]. In infection-related systems, SA can also identify the reactions and parameters that contribute most strongly to the system dynamics [22].

Variance-based global sensitivity analysis (GSA) using the Sobol' method decomposes the variance of a model output into contributions from individual inputs and their interactions. The first-order index measures the main effect of an input, whereas higher-order indices quantify interactions among multiple inputs. The total-order index captures the main effect and all interaction effects involving a particular input. For independent inputs, a total-order index greater than the corresponding first-order index indicates a contribution from interactions involving that input. Similar values indicate that interaction effects are comparatively small.

The analysis involves defining the input parameter ranges, generating samples using Sobol sequences, evaluating the model for the sampled inputs, and calculating the sensitivity indices. Independent uniform distributions can be assigned to the uncertain inputs over their specified lower and upper bounds.

Global sensitivity analysis distinguishes uncertainty in initial state values from uncertainty in kinetic parameters, as these represent different sources of model uncertainty. A high Sobol' index for an initial state value indicates a substantial contribution to output variance within the specified input ranges, but does not necessarily imply dominance of the associated biochemical mechanism. Thus, Sobol' indices reflect model-output sensitivity under the adopted model structure and sampling assumptions, rather than direct evidence of mechanistic dominance.

4.1 First-Order Sensitivity Index

The first-order Sobol' index quantifies the fraction of output variance attributable to the main effect of an individual input. Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ denote the vector of n independent uncertain inputs, and let Y denote a scalar model output. The first-order index for X_i is defined as

$$S_i = 1 - \frac{\mathbb{E}_{X_i}\left[\mathrm{Var}_{\mathbf{X}_{\sim i}}(Y \mid X_i)\right]}{\mathrm{Var}(Y)}. \tag{27}$$

Here, $\mathbf{X}_{\sim i}$ represents all inputs except X_i, $\mathbb{E}$ denotes expectation, and Var denotes variance. A larger S_i indicates a greater contribution of the individual input to the output variance, averaged over the distributions of the remaining inputs.

4.2 Total-Order Sensitivity Index

The total-order Sobol' index, denoted by S_{T_i}, measures the contribution of X_i to output variance, including all interactions involving that input. It is defined as

$$S_{T_i} = 1 - \frac{\mathrm{Var}_{\mathbf{X}_{\sim i}}\left(\mathbb{E}_{X_i}[Y \mid \mathbf{X}_{\sim i}]\right)}{\mathrm{Var}(Y)}. \tag{28}$$

5 Results

5.1 Time-course profiles of the model state variables

The simulations examine the temporal dynamics of the kallikrein–kinin system (KKS), the terminal complement pathway, and the associated inflammatory response under controlled and uncontrolled conditions. Additional perturbation simulations assess the effects of two-fold increases and two-fold decreases in the initial values of selected state variables, including BK, B2R, P-KAL, KAL, HK, C5a, and C5aR1. The analysis focuses on BK, the BK–B2R complex, the C5a–C5aR1 complex, IL-6, and the model-defined states associated with icatibant and vilobelimab (IFX-1). Figures 4–8 present the corresponding simulation results. Throughout this section, the state-variable values

Variables	Entities	**Current value**	**Lower limit**	**Upper limit**
x_1	BK	3.0463	2.7417	3.3510
x_2	B2R	4.1555	3.7399	4.5711
x_3	BK-B2R complex	0.000	0.0000	1.000
x_4	TAFIa	3.3417E-6	3.0075E-6	3.6759E-6
x_5	P-KAL	3.2899	2.9609	3.6189
x_6	KAL	4.8586	4.3727	5.3445
x_7	HK	2.7339	2.4606	3.0074
x_8	P-KAL-HK complex	0.0000	0.0000	1.0000
x_9	IL-6	3.5784	3.2206	3.9362
x_{11}	C5	3.8597	3.4737	4.2457
x_{12}	C5a	2.5732	2.3159	2.8305
x_{13}	C5b	1.2865	1.1579	1.4152
x_{14}	C5aR1	10.1662	9.1496	11.1829
x_{15}	C5a-C5aR1 complex	0.0000	0.0000	1.0000

Table 1: **Model state-variable reference values and ranges used in global sensitivity analysis (GSA).** The table lists the reference values assigned to the model state variables and the corresponding lower and upper bounds used to define uniform input distributions for variance-based GSA using the Sobol' method. These values represent model-defined abundances in the adopted numerical formulation rather than experimentally calibrated biochemical concentrations and should not be interpreted as absolute molar concentrations.

represent model-defined numerical abundances rather than experimentally calibrated biochemical concentrations.

5.1.1 Bradykinin and the BK–B2R complex

Figure 4 compares the simulated dynamics of BK and the BK–B2R complex under controlled and uncontrolled conditions. Under icatibant-related control, the BK trajectory exhibits a pronounced transient disturbance before declining towards a low level (Fig. 4A). Without control, BK initially decreases and subsequently increases, reaching a peak of approximately 4.26 model-defined units after approximately 80 s

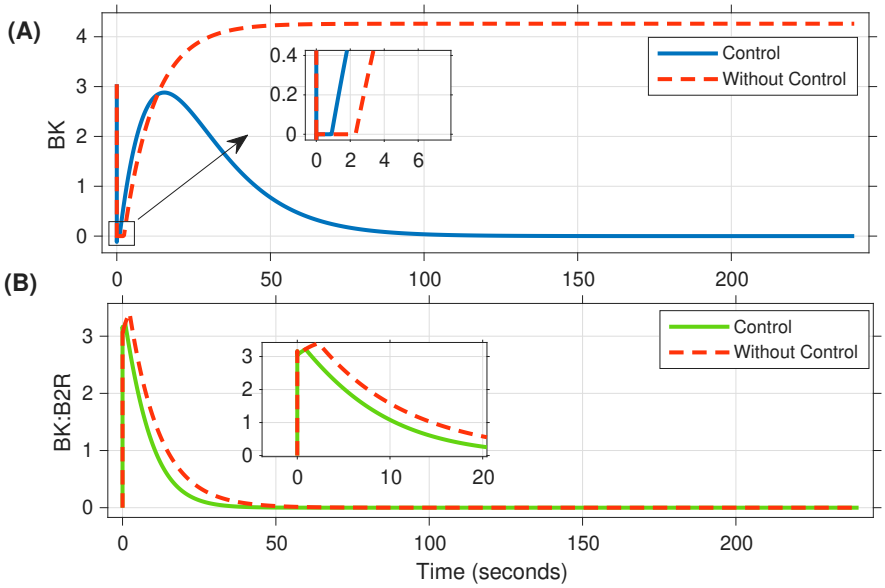


Fig. 4: Simulated dynamics of (A) bradykinin (BK) and (B) the BK–B2R complex under controlled and uncontrolled conditions. The trajectories illustrate the simulated response to the icatibant-related control input. The plotted values represent model-defined state variables rather than experimentally calibrated biochemical concentrations.

(Fig. 4A). The contrasting trajectories demonstrate the effect of the control input on the simulated BK response.

The BK–B2R complex exhibits a transient increase, reaching a peak value of approximately 3.16 model-defined units before declining towards its initial value of zero (Fig. 4B). Within the model, the formation of the P-KAL–HK complex contributes to the dynamics of the KKS and influences the simulated BK response. The GSA further identifies B2R as an influential input for BK, whereas BK and B2R make substantial contributions to the variance of the BK–B2R complex (Fig. 9A,B).

Figure 5 illustrates the effects of initial-condition perturbations on the BK–B2R complex. Two-fold increases and decreases in the initial values of BK, B2R, P-KAL, KAL, and HK produce distinct trajectories and peak values. These differences demonstrate the sensitivity of the simulated complex dynamics to changes in the initial conditions of the selected KKS components. Overall, the simulations show that the modelled BK response depends on the interactions among pathway components and the application of the icatibant-related control input.

5.1.2 C5a–C5aR1 complex dynamics

Figure 6 presents the simulated dynamics of the C5a–C5aR1 complex. In the mathematical model, KAL participates in the cleavage of complement component C5

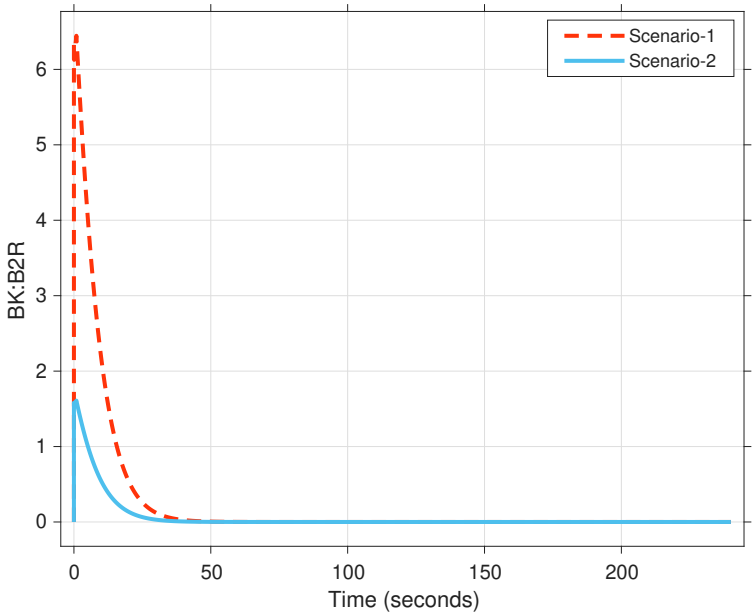


Fig. 5: Simulated dynamics of the BK–B2R complex under two initial-condition perturbation scenarios. Scenario 1 represents a two-fold increase, whereas Scenario 2 represents a two-fold decrease in the initial values of the selected model state variables. The trajectories illustrate the sensitivity of the simulated complex dynamics to perturbations in the model-defined numerical abundances.

to generate C5a. C5a subsequently interacts with C5aR1 to form the C5a–C5aR1 complex, which contributes to the modelled IL-6 response.

Under the controlled condition, the C5a–C5aR1 complex reaches a peak value of approximately 2.57446 model-defined units (Fig. 6A). The vilobelimab-related control input modifies the simulated complex trajectory, indicating a regulatory effect within the adopted mathematical framework. The perturbation simulations produce peak values of approximately 5.14893 and 1.28723 model-defined units under the two-fold increase and decrease scenarios, respectively (Fig. 6B). These results demonstrate that changes in the initial values of the selected state variables substantially alter the simulated complex response.

The GSA identifies C5a as the dominant contributor to variations in the C5a–C5aR1 complex, while C5aR1 exhibits a moderate influence within the selected input ranges (Fig. 9C). These findings highlight the roles of complement activation and receptor-associated interactions in determining the simulated dynamics of the complex.

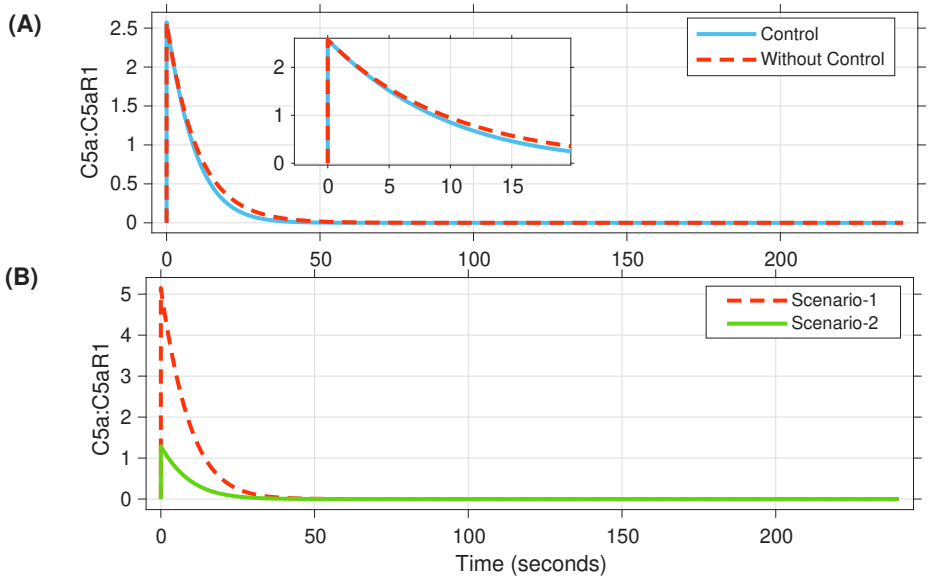


Fig. 6: Simulated dynamics of the C5a–C5aR1 complex (A) under controlled and uncontrolled conditions and (B) under two initial-condition perturbation scenarios. Scenario 1 represents a two-fold increase, whereas Scenario 2 represents a two-fold decrease in the initial values of the selected model state variables. The trajectories illustrate the effects of initial-condition perturbations and the vilobelimab-related control input on the simulated complex dynamics. The plotted values represent model-defined numerical abundances rather than experimentally calibrated biochemical concentrations.

5.1.3 Interleukin-6 dynamics

Figure 7 compares the simulated IL-6 response under controlled and uncontrolled conditions. Under control, IL-6 reaches a peak value of approximately 8.4011 model-defined units before approaching a steady-state value of 6.87369. Without control, the peak increases to approximately 9.84554, followed by stabilization at 8.16235 (Fig. 7A). Relative to the uncontrolled steady-state value, the controlled trajectory shows an approximately 15.8% reduction in the model-defined IL-6 state.

Figure 7B illustrates the effects of initial-condition perturbations on the IL-6 response. A two-fold increase in the initial values of C5a, C5aR1, and KAL produces a peak value of approximately 13.3468 model-defined units, followed by stabilization at 10.9363. Conversely, a two-fold decrease in these initial values produces a peak of approximately 5.92112 and a steady-state value of 4.84397. These results demonstrate that the initial conditions of the selected complement-related components influence the magnitude of the simulated IL-6 response.

The GSA identifies the BK–B2R and C5a–C5aR1 complexes as influential contributors to variations in the model-defined IL-6 state (Fig. 9D). B2R also exhibits a substantial influence, whereas BK and C5a show comparatively smaller effects

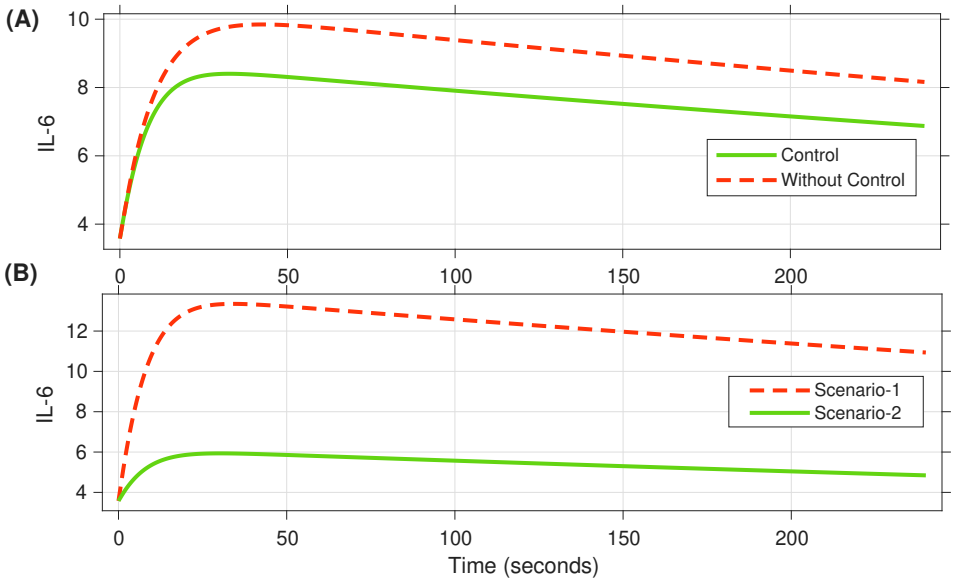


Fig. 7: Simulated dynamics of interleukin-6 (IL-6) (A) under controlled and uncontrolled conditions and (B) under two initial-condition perturbation scenarios. Scenario 1 represents a two-fold increase, whereas Scenario 2 represents a two-fold decrease in the initial values of the selected model state variables. The trajectories illustrate the effects of control and initial-condition perturbations on the simulated IL-6 response. The plotted values represent model-defined numerical abundances rather than experimentally calibrated biochemical concentrations.

in the reported analysis. Taken together, the controlled and perturbed simulations demonstrate that the model-defined IL-6 response depends on both the control inputs and the initial conditions of interacting pathway components. These numerical changes describe simulated model behaviour and do not directly quantify changes in experimentally measured IL-6 concentrations.

5.1.4 Control performance of icatibant and vilobelimab

Figure 8 evaluates the reference-tracking performance of the proposed DISMC for the model-defined states associated with icatibant and vilobelimab (IFX-1), together with their corresponding control inputs. The icatibant-related state approaches its reference value of 2 in the adopted numerical formulation (Fig. 8A). Its corresponding control input reaches a maximum value of approximately 1.2 before gradually decreasing towards zero (Fig. 8B).

Similarly, the vilobelimab-related state converges towards its reference value of 3 (Fig. 8C). The associated control input reaches a maximum value of approximately 3.736 before decreasing towards zero (Fig. 8D). These trajectories demonstrate the controller's reference-tracking behaviour and the evolution of the control effort over the simulated time interval.

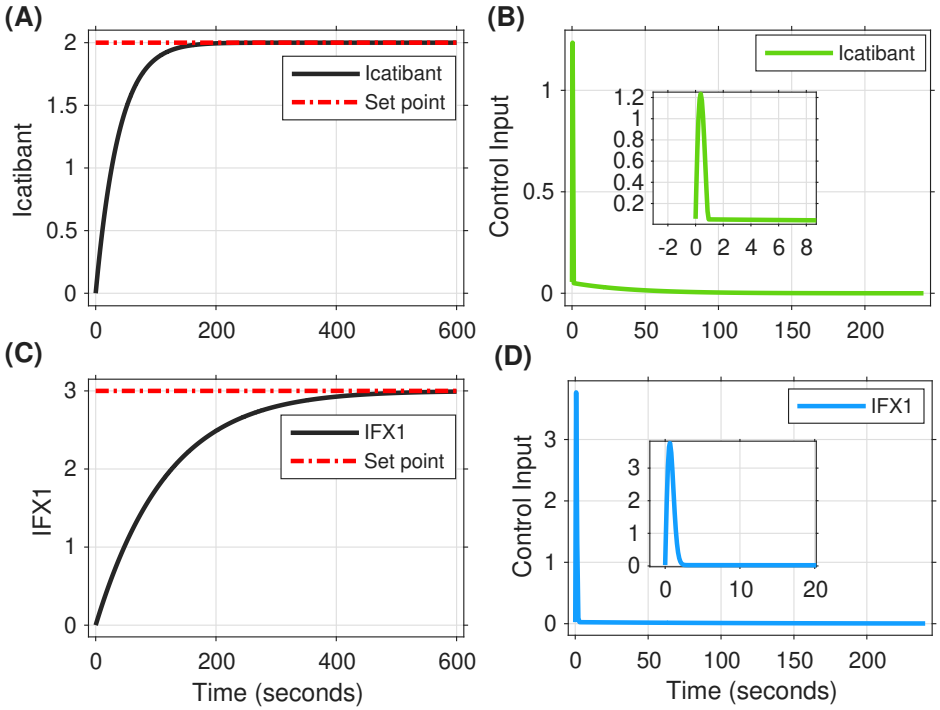


Fig. 8: Simulated control performance of the double-integral sliding mode controller (DISMC) for the model-defined states associated with (A) icatibant and (C) vilobelimab (IFX-1), together with their corresponding control inputs in (B) and (D), respectively. The controller regulates the icatibant-related and vilobelimab-related states towards their reference values of 2 and 3, respectively. The trajectories illustrate reference tracking and the temporal evolution of the corresponding control inputs. The reference values represent model-defined control set points rather than experimentally calibrated drug concentrations, administered doses, or clinically validated therapeutic targets.

The results demonstrate that the DISMC regulates both drug-associated model states towards their specified reference values within the adopted mathematical framework. However, the reference values and control-input magnitudes represent model-defined numerical quantities. They do not establish experimentally measured drug concentrations, administered doses, or clinically validated treatment regimens.

5.2 Variance-based global sensitivity analysis

The variance-based global sensitivity analysis (GSA) uses the Sobol' method to assess the contributions of selected model inputs to variations in BK, the BK–B2R complex, the C5a–C5aR1 complex, and IL-6 (Fig. 9). In addition, Fig. 10 presents the effects of kinetic-parameter variations on IL-6. First-order Sobol' indices quantify the individual

contributions of inputs to output variance, whereas total-order indices account for both individual effects and interactions among inputs.

For BK, B2R is the most influential input, followed by P-KAL and the P-KAL–HK complex (Fig. 9A). These results indicate that the simulated BK response is sensitive to variations in receptor-related state values and components involved in KKS activation.

For the BK–B2R complex, BK and B2R are the dominant contributors to output variance (Fig. 9B), consistent with their direct roles in complex formation. The remaining inputs have smaller effects within the specified sensitivity-analysis ranges.

For the C5a–C5aR1 complex, C5a is the dominant input, while C5aR1 makes a moderate contribution (Fig. 9C). This pattern indicates that the simulated complex response is sensitive to variations in the ligand and receptor state values.

For IL-6, the C5a–C5aR1 complex is a major contributor to output variance, alongside substantial contributions from the BK–B2R complex and B2R (Fig. 9D). BK and C5a have comparatively smaller effects, while the remaining inputs contribute less to the reported analysis. These findings indicate that the simulated IL-6 response is sensitive to the coupled dynamics of the KKS and complement pathways.

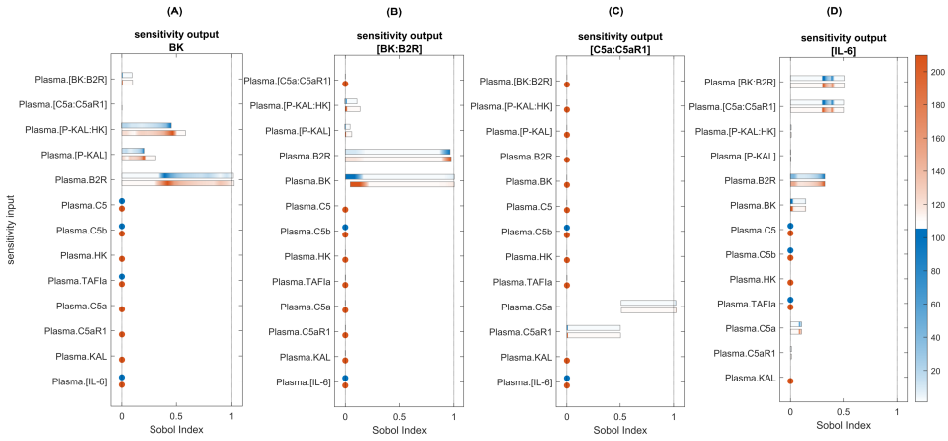


Fig. 9: **Variance-based global sensitivity analysis using the Sobol' method.** Bar plots show the first-order and total-order Sobol' indices for (A) BK, (B) the BK–B2R complex, (C) the C5a–C5aR1 complex, and (D) IL-6. First-order indices quantify the individual contributions of the model inputs to output variance, whereas total-order indices account for both individual effects and interactions. A total-order index exceeding its corresponding first-order index indicates a contribution from interactions with other inputs. (A) B2R has the largest influence on BK, with additional contributions from P-KAL and the P-KAL–HK complex. (B) BK and B2R dominate the variance of the BK–B2R complex. (C) C5a is the principal contributor to the variance of the C5a–C5aR1 complex, while C5aR1 has a moderate influence. (D) The C5a–C5aR1 complex is a major contributor to IL-6 variance, with additional contributions from the BK–B2R complex and B2R. The results characterise the sensitivity of the model outputs to variations in the selected inputs within the specified sampling ranges.

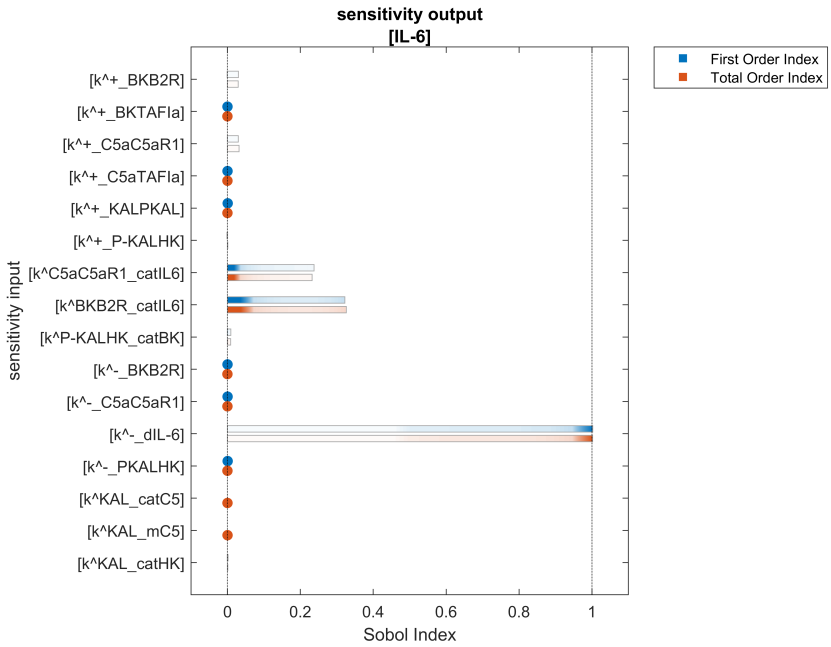


Fig. 10: First-order and total-order Sobol sensitivity indices for IL-6 concentration. The blue and orange markers represent the first-order and total-order sensitivity indices, respectively, while the horizontal bars indicate the corresponding sensitivity estimates. The results identify $[k^{-}_{d\mathrm{IL6}}[$ as the dominant sensitivity input, with both indices approaching 1, indicating that this parameter accounts for most of the variance in IL-6 output. The interaction terms $[k^{C5aC5aR1}_{catIL6}]$ and $[k^{BKB2R}_{catIL6}]$ exhibit comparatively smaller contributions, with the latter showing a higher sensitivity index. The remaining parameters have negligible sensitivity indices, suggesting limited contributions to the output variance under the specified model and parameter distributions.

The kinetic-parameter analysis in Fig. 10 complements the state-variable analysis by examining how uncertainty in the model's kinetic parameters contributes to variations in IL-6.

The resulting sensitivity rankings depend on the model structure, the ranges assigned to the inputs, and the sampling assumptions. Total-order indices exceeding their corresponding first-order indices indicate contributions from interactions among inputs. Moreover, a high sensitivity index for an initial state value indicates that uncertainty in that value contributes substantially to output variance; it does not, by itself, establish the dominance of the associated biochemical mechanism. Thus, the GSA quantifies the sensitivity of simulated outputs to model inputs rather than directly demonstrating biological or mechanistic importance.

6 Discussion

This study presents a mathematical modelling and nonlinear control framework for investigating the coupled inflammatory dynamics of the kallikrein–kinin system (KKS) and terminal complement pathway in SARS-CoV-2 infection. The model incorporates key mediators, including bradykinin (BK), the bradykinin B2 receptor (B2R), prekallikrein (P-KAL), high-molecular-weight kininogen (HK), C5a, C5a receptor 1 (C5aR1), and interleukin-6 (IL-6). A double-integral sliding mode controller (DISMC) was developed to investigate the regulation of the BK–B2R and C5a–C5aR1 pathways through hypothetical control inputs associated with icatibant and vilobelimab (IFX-1), respectively. By integrating biochemical pathway modelling, genetic algorithm (GA)-based parameter optimisation, Lyapunov analysis, and variance-based global sensitivity analysis (GSA), this study evaluates the behaviour of the model under specified simulation conditions.

Previous studies have employed mathematical models to investigate SARS-CoV-2 infection dynamics, host immune responses, and potential treatment strategies. These include models of viral interactions with epithelial cells [23], antibody- and cytotoxic T lymphocyte-mediated immune responses [24], and optimal control based on Pontryagin's maximum principle [25]. In contrast, the present study focuses on the computational regulation of interconnected inflammatory pathways. Its principal contribution is the application of DISMC to examine the coordinated dynamics of the KKS and terminal complement pathway and their relationship with the modelled IL-6 response.

The uncontrolled simulations demonstrate interactions among BK, B2R, P-KAL, HK, and their associated complexes, with distinct transient and steady-state behaviours. Previous studies have discussed the potential involvement of dysregulated kallikrein–kinin signalling in COVID-19-associated inflammation [16, 26]. In the present model, the elevated BK state is accompanied by changes in BK–B2R complex dynamics and an elevated IL-6 state, reflecting the relationships encoded in the biochemical formulation. These observations are consistent with discussions of the potential role of dysregulated bradykinin signalling in COVID-19 [16, 27]. However, the simulated relationships do not independently establish the magnitude or causality of BK–B2R-mediated inflammation in patients.

Similarly, the terminal complement pathway model represents C5 cleavage, C5a interaction with C5aR1, and the resulting dynamics of the C5a–C5aR1 complex. Changes in these states influence the modelled IL-6 response. This relationship is consistent with the findings of Carvelli et al. [3], who reported an association between C5a–C5aR1 signalling, IL-6 responses, and inflammatory severity in COVID-19. These results illustrate the computational value of representing complement-mediated signalling alongside the KKS within a coupled mathematical framework, although they do not independently establish biological causality or clinical disease severity.

The icatibant-associated control input modifies the simulated BK-related pathway and demonstrates reference tracking under DISMC. The corresponding model-defined reference value is 2, which represents a numerical control set point rather than an experimentally calibrated drug concentration, administered dose, or validated therapeutic target. Perturbations in BK, B2R, P-KAL, and HK further demonstrate that

the simulated pathway response depends on multiple interacting components and their initial conditions. Similarly, the vilobelimab-associated control input demonstrates reference tracking towards a model-defined reference value of 3. Together, these control channels illustrate the mathematical feasibility of regulating selected pathway states under the adopted simulation assumptions. However, these findings do not establish the clinical effectiveness, safety, or optimal dosing of icatibant or vilobelimab.

A key quantitative result is the reduction in the simulated IL-6 state from 8.16 in the uncontrolled scenario to 6.87 in the controlled scenario, corresponding to an approximately 15.8% decrease relative to the uncontrolled value. This result indicates that DISMC can reduce the modelled IL-6 state under the specified simulation conditions. Nevertheless, because the state variables represent model-defined abundances rather than experimentally calibrated biochemical concentrations, this percentage should not be interpreted as a measured reduction in IL-6 in patients or as evidence that the controller prevents cytokine storm or improves clinical outcomes.

Variance-based GSA using the Sobol' method identified BK and B2R as influential inputs for the BK–B2R complex, while C5a and C5aR1 substantially influenced the C5a–C5aR1 complex. These complexes also contributed to variations in the simulated IL-6 response. Two-fold increases and decreases in selected initial state values further demonstrated that perturbations can alter pathway trajectories and control responses. These findings help identify influential variables within the mathematical formulation. However, sensitivity rankings depend on the model equations, selected input ranges and probability distributions, and output definitions. They should therefore be interpreted as measures of model-output sensitivity rather than definitive rankings of biological importance or mechanistic dominance in patients.

Although the Sobol'-based global sensitivity analysis identifies kinetic parameters that contribute to variations in the simulated model outputs, it does not directly establish the robustness of the closed-loop double-integral sliding mode controller (DISMC) to uncertainty in these parameters. In particular, the present analysis does not quantify whether acceptable tracking performance is maintained when estimated kinetic parameters vary substantially while the controller gains remain fixed at their nominal, genetic algorithm-optimised values. A dedicated closed-loop parameter-variation study would be required to evaluate this aspect quantitatively. Therefore, assessing the robustness of DISMC to large kinetic-parameter variations remains an important direction for future investigation.

GA-based optimisation and Lyapunov analysis provided complementary tools for controller design and evaluation. The optimisation procedure was used to select controller parameters, while the Lyapunov analysis examined the reaching behaviour of the sliding variables under the nominal reaching law. However, reaching of the sliding variables does not, by itself, establish stability of the complete biochemical system or convergence of every individual tracking error. Moreover, GA-based optimisation does not guarantee a globally optimal solution without appropriate global optimality conditions. The current analysis also does not establish guaranteed robustness against external disturbances or parameter uncertainties, as these effects were not explicitly formulated with defined bounds in the model.

Several limitations should be considered when interpreting these findings. First, the biochemical network is represented by deterministic ordinary differential equations (ODEs), and its state variables have not been calibrated against experimentally measured absolute protein concentrations. Although microarray gene-expression data [1] provide biological context for model development, their correspondence with the modelled biochemical states has not been established. Future work should incorporate experimental protein and cytokine measurements, parameter calibration, and independent validation to improve the biological interpretability and predictive capability of the framework.

Second, the icatibant- and vilobelimab-associated control inputs are hypothetical mathematical signals rather than clinical doses. The model does not incorporate pharmacokinetic/pharmacodynamic (PK/PD) relationships, drug exposure, or clinically relevant treatment constraints. Future extensions should integrate validated PK/PD relationships and bounded, non-negative control inputs to support more realistic investigations of therapeutic optimisation.

Third, the discontinuous sign function in the controller may produce chattering, and no explicit boundary-layer or smoothing mechanism has been implemented. Furthermore, the present stability analysis does not establish stability of the complete biochemical system or robustness under bounded disturbances and parameter uncertainties. Future studies should investigate positivity and boundedness of the relevant states, robust or adaptive control strategies, tracking accuracy, and control effort.

Fourth, the GSA results depend on the assumed input ranges, probability distributions, model structure, and output definitions. Further assessment of sensitivity-index convergence and parameter uncertainty, alongside stochastic modelling, could help characterise the effects of biological variability. For example, Ghosh et al. [28] investigated inherited randomness in a stochastic model of leprosy, illustrating the use of stochastic formulations in mathematical disease modelling. Such approaches may inform future extensions of the present framework, although their applicability to inflammatory biochemical pathways requires separate evaluation.

Alternative intervention schedules and therapeutic optimisation strategies also warrant investigation. Cao et al. [29] explored optimal control in a mathematical model of leprosy, while Mukherjee et al. [30] investigated impulsive control for stem-cell-based treatment in leprosy pathogenesis. Similarly, Bag et al. [31] developed a mathematical framework for examining chemo-iPSC therapy in cervical cancer. These studies provide methodological examples of treatment optimisation and intervention modelling in disease-related systems. Beyond biomedical applications, Ahammed et al. [32] investigated a mathematical framework for sterile insect technique (SIT) in pest management, illustrating the broader application of dynamical modelling to intervention strategies. Although these studies address different biological systems and objectives, their methodological approaches may inform future developments following appropriate adaptation to the inflammatory pathways considered here.

Overall, this study provides a computational proof of concept for applying DISMC to a coupled KKS–terminal complement model of SARS-CoV-2-associated inflammation. The simulations demonstrate reference tracking of the two modelled control

channels, changes in pathway-complex dynamics, and a reduction in the modelled IL-6 response. GSA identifies influential model inputs, while optimisation and Lyapunov analysis support controller design and the assessment of nominal reaching behaviour. Experimental calibration, independent validation, realistic treatment constraints, and systematic evaluation of positivity, stability, uncertainty, and robustness remain essential before the framework can support quantitative biological predictions or clinical applications.

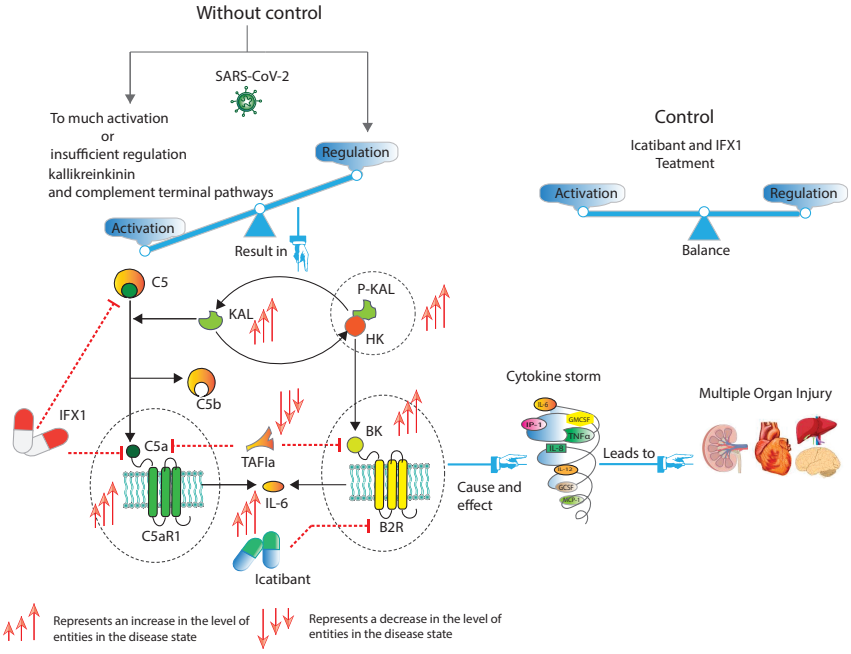


Fig. 11: **The study's implications.** In the uncontrolled disease state, impaired pathways—resulting from imbalanced activation and regulation of key components—lead to hyperactivation of mediators such as pre-kallikrein (P-KAL), kallikrein (KAL), high-molecular-weight kininogen (HK), bradykinin (BK), the bradykinin B2 receptor (B2R), potent complement anaphylatoxin (C5a), complement component 5a receptor 1 (C5aR1) and their downstream products (e.g., BK-B2R and C5a-C5aR1). Concurrently, the inhibitor TAFIa exhibits suppressed activity. Pharmacological intervention with IFX1 (a C5/C5a inhibitor) and icatibant (a B2R antagonist) restores systemic balance by countering these dysregulations, steering the inflammatory response toward homeostasis and improving survival outcomes.

7 Conclusion

This study presents a computational framework integrating a mathematical model of the kallikrein–kinin system (KKS) and the terminal complement pathway with a double-integral sliding mode controller (DISMC) to investigate the regulation of SARS-CoV-2-associated inflammatory dynamics. Genetic algorithm-based optimisation was used to select controller parameters, while Lyapunov analysis established the nominal reaching behaviour of the sliding variables. Numerical simulations examined the controlled and uncontrolled dynamics of bradykinin (BK), the BK–B2R complex, the C5a–C5aR1 complex, and interleukin-6 (IL-6), as well as the effects of perturbations in selected initial-state values. Under the simulated conditions, the controlled IL-6 trajectory exhibited an approximately 15.8% reduction in its steady-state model-defined value relative to the uncontrolled case. The drug-associated model states also tracked their prescribed reference values, demonstrating the operation of the proposed control framework within the adopted mathematical formulation. Sobol'-based global sensitivity analysis identified influential initial-state values and kinetic parameters contributing to variations in the selected outputs, providing complementary information about model sensitivity. Nevertheless, convergence of the sliding variables does not, by itself, establish convergence of every individual tracking error or stability of the complete biochemical system. Similarly, the present analysis does not establish guaranteed robustness to external disturbances or parameter uncertainties, nor does it demonstrate chattering reduction. Furthermore, the model states are not calibrated against experimentally measured absolute biochemical concentrations, and the drug-associated control inputs and reference values represent hypothetical mathematical quantities rather than clinically validated doses or therapeutic targets. Future work should prioritise experimental calibration and independent validation, incorporation of pharmacokinetic/pharmacodynamic relationships and clinically relevant treatment constraints, and systematic evaluation of state positivity, boundedness, tracking performance, stability, and robustness. Subject to these developments, the proposed framework may provide a foundation for further computational investigation of nonlinear control strategies in coupled inflammatory systems. Its present contribution is a mathematical proof of concept rather than evidence of clinical efficacy or an established drug-dosing strategy.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Supporting information

The Supporting Information (SI) includes: (I) Table S1, which lists the kinetic constants; (II) Figures S1 and S2, which illustrate Time-course profiles of the HK and P-KAL-HK in the disease state.

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